Dynamic Regime Transition Probabilities and Macro-Calibrated Portfolio Allocation

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Abstract

This research develops a probabilistic, data-driven framework for monthly regime forecasting and dynamic portfolio construction grounded in macro-financial transition dynamics. The model combines a time-varying transition probability (TVTP) system for macro regimes—*Goldilocks, Heating Up, Slow Growth, and Stagflation*—with monotone conditional "factor heads" capturing vulnerabilities in sentiment, liquidity, and labor markets. Each transition probability is conditioned on growth and inflation trends as well as regime duration, while the factor heads provide calibrated probabilities of structural weakness given current and forecasted macro conditions. Forward propagation uses a ridge-regularized VAR(1) to project key macro factors, enabling consistent one- and two-month-ahead regime distributions. Probabilities are calibrated through isotonic and Platt mappings to ensure reliability and interpretability. The resulting probability vectors serve as allocation signals in a dynamic portfolio strategy that tilts exposure toward the most probable economic regimes, constrained by trend filters and stability safeguards. Out-of-sample results (2020–2025) demonstrate high predictive accuracy, reliable probability calibration, and robust portfolio performance across equities, commodities, and fixed income. The approach unifies regime-based macro forecasting with machine-assisted portfolio design, offering a transparent bridge between statistical learning and practical asset allocation.

1 Macro overview

We model macro regime evolution with observable regimes and factor-driven sublabels to produce, for each month, full probability vectors for *next month* ($m1$) and *two months ahead* ($m2$) across all composite regimes (macro $\times$ weak flags), plus macro aggregates. The stack is:

- a **time-varying transition probability (TVTP)** macro layer for $P(m_{t+1} \mid m_t, X_t)$, where $m_t \in$ Goldilocks, HeatingUp, SlowGrowth, Stagflation is today's macro regime and $X_t = (G_t, I_t, d_t)$ are today's Growth/Inflation trend levels and the current regime duration
- three **monotone heads** for weak Sentiment, weak Liquidity/Credit, and weak Labor, giving $P(\text{flag} \mid m_{\text{dest}}, \text{factors})$, where $m_{\text{dest}} \in$ Goldilocks, HeatingUp, SlowGrowth, Stagflation is the destination macro and factors $= (S_t, \Delta S_t, , LQ_t, \Delta LQ_t, , LB_t, \Delta LB_t)$ are today's Sentiment/Liquidity/Labor trend levels and 1-month changes (for $m2$ use $t{+}1$ VAR forecasts in place of t)

- a **VAR(1)**: $Y_{t+1} = a + Y_t\Theta + \varepsilon_{t+1}$ for $Y_t = (G_t, I_t, S_t, LQ_t, LB_t)$, fit on *post-1990* data to avoid pre-1990 structural breaks (data imbalances) and reflect modern dynamics; *ridged* because we ridge-regularize coefficients $\hat{B} = (X^\top X + \lambda R)^{-1} X^\top Y$ (no penalty on the intercept[1]) to stabilize estimates and curb overfit, yielding $X_{t+1}^{\text{forecast}}$ for the $m2$ step (with path-wise duration update)

Data & labels. The macro regime $m_t \in$ {Goldilocks, Heating Up, Slow Growth, Stagflation} is determined by the directions of `GrowthIndex_Trend_xt` and `InflationTrend_xt`. The composite label (`EconRegimSpec`) appends *weakSentiment / weakLiquidity / weakLabor* based on the levels and short-run changes of `SentimentIndex_Trend_xt`, `LiquidityCreditIndex_Trend_xt`, and `LaborIndex_Trend_xt`. Duration is the number of consecutive months spent in m_t.

2 Method

2.1 TVTP macro layer

We estimate

$$P(m_{t+1} = j \mid m_t = i, X_t) = \frac{\exp\{\beta_{j,0} + \beta_{j,G} G_t + \beta_{j,I} I_t + s_j(d_t)\}}{\sum_k \exp\{\beta_{k,0} + \beta_{k,G} G_t + \beta_{k,I} I_t + s_k(d_t)\}},$$

where G_t and I_t are Growth/Inflation trends, d_t is duration in the current macro, and $s_j(\cdot)$ is a smooth duration effect[2]. We calibrate probabilities for reliability out-of-sample[3].

2.2 Monotone factor heads

For each destination macro m, and for each flag $f \in$ {weakSentiment, weakLiquidity, weakLabor}, we estimate

$$P(f_{t+1} = 1 \mid m_{\text{dest}} = m, \text{factor level}, \Delta\text{factor}),$$

with monotone intent (deteriorating levels $\Rightarrow$ higher weak probability), probability calibration, and empirical-Bayes shrinkage toward macro-conditional base rates to stabilize sparse slices[4].

2.3 Two-step propagation (VAR + duration paths)

To form $m2$, we must carry information through $t+1$:

[1] No penalty on the intercept: shrinking a would distort the unconditional mean and induce level bias; ridge acts only on dynamic coefficients so $E[Y_{t+1}] = a + E[Y_t]\Theta$ remains properly calibrated, especially when regressors are not centered. Regressors are left uncentered because we work in real time on raw index scales. Demeaning would require window/rolling means (risking look-ahead and regime-drift instability) and would shift the intercept, whereas an unpenalized intercept cleanly absorbs the level. Ridge on dynamic coefficients provides stability without altering level calibration.

[2] $s_j(d_t) = \sum_{r=1}^{R} \gamma_{jr} B_r(d_t)$ is a cubic B-spline of duration with knots at empirical duration quantiles (e.g., 4–6 interior knots) to capture nonlinearity

[3] After fitting the multinomial TVTP we calibrate class probabilities via classwise *isotonic regression* (a type of regression analysis that finds a non-decreasing (or non-increasing) function that best fits a dataset, minimizing the mean squared error): monotone mapping on a pre-2020 holdout (all observations with Date $\leq$ 2019-12 (1960–2019), reserved for probability calibration (isotonic/Platt) and tuning, while 2020+ is kept strictly for out-of-sample evaluation), CV-smoothed (K-fold, e.g., 5-fold, isotonic calibration using out-of-fold scores, then averaging/refitting the monotone map across folds to reduce variance and avoid overfitting.), with *Platt scaling* $p = \sigma(ascore + b)$ as fallback when isotonic is unstable or positives are scarce.

[4] For each head and destination macro m, shrink the raw probability $\hat{p}$ toward the macro–conditional base rate $\bar{p}_m$ via a Beta–Binomial prior $a = k\bar{p}_m,\ b = k(1-\bar{p}_m)$, yielding $\tilde{p} = \frac{a+y}{a+b+n} \approx \frac{k\bar{p}_m + n\hat{p}}{k+n}$ (with n samples, y positives); k (e.g., 30 for Sent/Liq, 60 for Labor) sets shrink strength, stabilizing sparse slices and avoiding overconfident extremes.

1. Forecast the five trends (G, I, S, LQ, LB) with a ridged **VAR(1)** on a modern window (tuned and fixed at window start 1990, ridge $\lambda = 0.05$[5]).
2. Update duration path-wise: if $m_{t+1} = m_t$ then $d_{t+1} = d_t{+}1$, else $d_{t+1} = 1$.
3. Evaluate $P(m_{t+2} \mid m_{t+1}, X_{t+1}^{\text{forecast}}, d_{t+1})$ and sum over m_{t+1}.
4. Evaluate the heads at $t{+}2$ using the $t{+}1$ factor forecasts.

Why this approach. Regime changes are not memoryless and depend on prevailing macro drivers. TVTP captures state- and signal-dependent transitions; heads capture factor-conditioned vulnerabilities; the VAR provides coherent one-step factor forecasts for $m2$. Calibration, shrinkage, and duration deliver stability and interpretability.

3 Turning states into probability vectors (2020–2025)

For month t, the current state is $(m_t,\, G_t, I_t, S_t, LQ_t, LB_t,\, \Delta S_t, \Delta LQ_t, \Delta LB_t,\, d_t)$.

3.1 Next month ($m1$): composites and macro

Let a composite label be $c = (m, \mathcal{F})$, where $\mathcal{F}$ is the set of weak flags present in the label. Define the unnormalized mass

$$\widetilde{P}_{t+1}(c) \;=\; P(m_{t+1} = m \mid m_t, X_t) \;\times \prod_{f \in \{\text{weakS, weakLQ, weakLB}\}} \begin{cases} P(f_{t+1} = 1 \mid m, \text{factors}_t), & f \in \mathcal{F}, \\ 1 - P(f_{t+1} = 1 \mid m, \text{factors}_t), & f \notin \mathcal{F}. \end{cases}$$

Within each macro m, renormalize to match the macro mass exactly[6]:

$$P_{t+1}(c) \;=\; \frac{P(m_{t+1} = m \mid m_t, X_t)}{\sum_{c' :\, \text{macro}(c') = m} \widetilde{P}_{t+1}(c')}\; \widetilde{P}_{t+1}(c).$$

Macro probabilities are then sums over composites: $P_{t+1}(m) = \sum_{\text{macro}(c) = m} P_{t+1}(c)$.

3.2 Two months ($m2$): composites and macro

First form the macro distribution at $t{+}2$ by summing over paths m_{t+1}:

$$P_2(m) \;=\; \sum_{m_1} P(m_{t+1} = m_1 \mid m_t, X_t)\; P\big(m_{t+2} = m \mid m_{t+1} = m_1,\, X_{t+1}^{\text{VAR}},\, d_{t+1}^{\text{path}}\big).$$

[5] Ridge $\lambda = 0.05$ was selected by grid-tuning on a pre-2020 validation slice (2010–2019) to minimize the $m2$ log-loss; it provides enough shrinkage to stabilize the VAR(1) without oversmoothing dynamics, with sensitivity showing a shallow optimum around 0.05 (robust for $\lambda \in [0.02, 0.08]$).

[6] The within–macro renormalization enforces mass conservation so that composites under macro m sum exactly to $P(m_{t+1} = m)$ from the TVTP; it also corrects for (i) missing/omitted composite labels and (ii) small calibration errors in the heads that would otherwise leak or deficit probability within m.

Then compose composites using heads evaluated at $t+2$ (based on the $t+1$ factor forecasts) and renormalize within m exactly as above:

$$\widetilde{P}_{t+2}(c) \;=\; P_2(m) \;\times \prod_f \big[\text{flag prob at } t+2\big], \qquad P_{t+2}(c) \;=\; \frac{P_2(m)}{\sum_{c':\,\text{macro}(c')=m} \widetilde{P}_{t+2}(c')}\,\widetilde{P}_{t+2}(c),$$

and $P_{t+2}(m) = \sum_{\text{macro}(c)=m} P_{t+2}(c)$.

3.3 Deliverables

For every month t from 2020 onward we produce two wide matrices:

- $m1$: a row vector $P_{t+1}(c)$ over all composite labels c (sums to 1), plus macro aggregation $P_{t+1}(m)$.
- $m2$: a row vector $P_{t+2}(c)$ over all composite labels (sums to 1), plus macro aggregation $P_{t+2}(m)$.

These are saved as two Excel files (one for $m1$, one for $m2$), one row per month and one column per composite regime.

4 Performance (2020+ diagnostics)

- Macro-only TVTP: out-of-sample (OOS) log-loss ≈ 0.63; Top-1 accuracy ≈ 0.86[7].
- Composite $m1$: out-of-sample log-loss ≈ 1.50 (well below $\ln 18 \approx 2.89$ uniform baseline on frequent labels).
- Composite $m2$: out-of-sample log-loss ≈ 1.96 with VAR(1) tuned at window start 1990 and ridge 0.05.
- Sanity: head probabilities are calibrated and shrunk toward macro-conditional base rates; within each macro, composite probabilities renormalize to the macro mass exactly.

5 Robustness and safeguards

- **Structural change.** VAR fitted on a recent window with ridge shrinks instability; TVTP includes duration, allowing learned persistence/mean reversion.
- **Rare labels.** Evaluation focuses on composites with ≥ 12 occurrences to avoid noise; full distributions are still produced; empirical-Bayes shrinkage stabilizes sparse macro slices.
- **Missing data.** If a factor is missing at t, heads revert to macro-conditional base rates; the VAR imputes with training means for one-step forecasts only.
- **Interpretability.** Macro coefficients show how Growth/Inflation tilt transitions; heads reflect monotone vulnerability; any composite probability decomposes into macro step $\times$ flag step.
- **Monitoring.** Reliability curves/Brier scores, rolling OOS log-loss, and drift alerts (e.g., head probabilities > 0.9 or < 0.1 for extended periods) keep the system in check.

[7] Macro-only TVTP performance (2020+ OOS): log-loss ≈ 0.63 (lower is better: compared with the uniform baseline $\ln 4 \approx 1.386$) and Top-1 accuracy ≈ 0.86 (the highest-probability macro matches the realized macro in 86% of months).

6 How to use the vectors

- **Narrative.** "Given today's state, next-month macro probabilities are ...; two months out are ...; conditional weaknesses under each macro are ..."
- **Portfolio mapping.** Tilt exposure where $m2$ mass concentrates (turn the hogest-probability regimes into weights to overweight specific assets); overlay weak-flag odds (e.g., weakLiquidity) for risk-sensitive allocation.
- **Attribution.** Decompose spikes in a composite probability into macro persistence/rotation vs. factor-driven weakness via heads. Decompose a change in a composite $c = (m, \mathcal{F})$ into macro vs. head effects:

$$\Delta P(c) \approx \underbrace{\Delta P(m)\, H_t}_{\text{macro persistence/rotation via TVTP}} + \underbrace{P_t(m)\, \Delta H}_{\text{factor weakness via heads}}.$$

$$H_t = \prod_{f \in \mathcal{F}} q_t^{(m)}(f) \prod_{f \notin \mathcal{F}} \left[1 - q_t^{(m)}(f)\right], \qquad q_t^{(m)}(f) = \Pr(f_{t+1} = 1 \mid m,\ \text{factors}_t).$$

- **Sensitivity.** Compare baseline vs. counterfactual duration (reset $d_t = 1$) to quantify how time-in-state shifts path probabilities.

7 Dynamic Monthly Portfolio Construction (TVTP Regimes + Trend Filters)

We build a dynamic, monthly–rebalanced portfolio driven by time–varying transition probabilities (TVTP) over economic regimes. For each as–of month t, we read the regime probability vector $p_r(t)$ from the M1 file. For every regime r, we estimate each ticker's *adjusted return* as a Sharpe ratio using only history strictly prior to t and only rows where the resolved regime equals r: monthly *log* returns are annualized (mean/std). Tickers with fewer than `MIN_OBS_PER_REGIME` months in that regime are discarded; with `POSITIVE_ONLY` we also drop non–positive Sharpe.

Eligibility is then gated by trend filters measured at t. For indices/sectors/commodities we use the trend columns that literally contain "trend" in the Market and FI trend files; the rule is *positive and improving* (current > 0 and current $>$ prior by `TREND_DELTA_LAG`). For *stocks* we also require a price–trend test based on the exact `ticker_P` column in the Market trend file (this column name does not include the word "trend"); the default price rule is *improving or positive*. The trading universe is fixed:

Equities: `S5HLTH_P`, `S5ENRS_P`, `S5RLST_P`, `S5UTIL_P`, `S5CONS_P`, `SPW_P`, `RLV_P`, `SPX_P`, `S5INFT_P`, `S5TELS_P`, `S5COND_P`, `S5FINL_P`, `S5INDU_P`. *Commodities*: `CO1`, `HG1`, `XAUUSD`. *Fixed Income*: `UST1-3`, `UST20PLUS`, `UST3-5`, `UST5-7`, `UST3-10`, `UST5-10`, `UST3-7`, `USTALL`, `USAGG1-3`, `USAGG5-10`, `IG1-5`, `IG5-10`, `IG10PLUS`, `USHY`, `USHY1-5`, `HY0-5`, `MBS`.

If a regime r has no eligible tickers (e.g., sparse history or failed trends), we *do not* borrow across different roots. Instead, we walk *same–root fallbacks* by dropping modifiers stepwise (e.g., `root_a_b_c` $\to$ `root_a_b` $\to$ `root_a` $\to$ `root`) and use the closest fallback with eligibles. If even the root has none, that regime's probability is left *uninvested* (unless `REALLOCATE_MISSING` is enabled, which rescales invested sleeves to sum to one). We record the fraction actually deployed as *InvestedProb*.

Weights are formed *within* each (chosen) regime: its probability $p_r(t)$ is split across that regime's eligible tickers in proportion to $\text{Sharpe}^{\texttt{POWER}}$ (with `POWER`=1 by default; higher concentrates more in the best names). The portfolio for month $t+1$ uses weights computed at t. Realized portfolio return in $t+1$ is the weighted sum of *arithmetic* returns; we chain monthly P&L to produce Equity, CAGR, volatility, Sharpe, and max drawdown.

For analysis without rerunning the backtest, we save a *weights CSV* (as–of t) and then build a *monthly analytics* dataset by joining those weights to next–month returns and tagging each ticker to an asset class (Equities/Commodities/Fixed Income). From this we derive per–month class sleeves (weight, contribution, sleeve return), the total portfolio return, and *InvestedProb*; yearly aggregations power the interactive views: (i) yearly returns by asset class, (ii) yearly Sharpe by asset class, (iii) stacked average yearly weights (normalized to 100%) with class returns on a secondary axis, and (iv) per–year tables of the highest–return and longest–held tickers by asset class (computed over months actually held).